

CIXIO - Individual Summary

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What I Have Done

Initial Module Work – DevOps

- Worked on understanding the DevOps architecture of SmartHub.
 - Learned the purpose and usage of Docker, Docker Compose, Terraform, AWS infrastructure, and deployment workflows.
 - Successfully installed and validated Terraform locally.
 - Understood the role of Infrastructure as Code (IaC) and how Terraform can provision AWS resources such as VPCs, Subnets, Security Groups, and EC2 instances.
 - Studied how containerised services communicate within the SmartHub ecosystem.
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Knowledge Transfer & Learning Week

Day 1 (04/06/26 THU) – Team Discussions and Knowledge Sharing

- Participated in team discussions and knowledge-sharing sessions.
 - Shared my understanding of the DevOps module, including Docker, Terraform, deployment concepts, infrastructure provisioning, and containerised architectures.
 - Learned about the Authentication module from team members.
 - Understood JWT authentication, access and refresh tokens, bcrypt password hashing, role-based access, and CSV-based user onboarding.
 - Gained an initial understanding of the overall SmartHub architecture and how different modules interact.
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Day 2 (05/06/26 FRI) – LLM & RAG Deep Dive

- Studied the fundamentals of LLMs and how modern AI systems generate responses.
- Learned how Ollama, LLM models such as Llama 3.2 & Gemma3, embedding models, ChromaDB, and RAG pipelines work together.
- Set up Ollama locally and explored AI model management.
- Implemented a simple RAG workflow using PDF documents, Ollama, nomic-embed-text, and ChromaDB.
- Understood document ingestion, chunking, embeddings, vector storage, retrieval, and answer generation.
- Explored streaming responses and understood why they improve user experience.

Day 3 (06/06/26 SAT) – Exploring Alternatives & Model Comparison

- Compared Llama 3.2 with alternative models such as Gemma 3.
- Understood concepts such as hallucination and limitations of LLM reasoning.
- Tested Gemma 3 in a RAG setup and compared performance against Llama.
- Explored sentence-transformers as an alternative embedding framework.
- Studied pgvector and compared it with ChromaDB.
- Compared the pros and cons between dedicated vector databases and PostgreSQL-based vector storage.

Day 4 (07/06/26 SUN) – Mobile Module Learning

- Studied the SmartHub mobile architecture.
- Understood how Flutter enables Android and iOS development using a single codebase.
- Learned the purpose of widgets, screens, navigation, and Flutter project structure.
- Explored how mobile applications communicate with backend APIs.
- Understood Firebase Cloud Messaging (FCM), FCM token registration, and push notification delivery.
- Studied document uploads, chat functionality, offline caching, and user profile management from a mobile perspective.

Day 5 (08/06/26 MON) – DevOps Scaling & Infrastructure

- Studied how SmartHub can scale to support a large user base.
- Understood horizontal scaling, auto-scaling, Redis caching, RabbitMQ-based asynchronous processing, load balancing, and multi-instance deployments.
- Learned the purpose of AWS EC2 and how multiple EC2 instances can be placed behind a load balancer.
- Explored AWS S3 and its role in scalable file storage.
- Compared RabbitMQ with Apache Kafka and why Kafka is preferred on a larger scale.

Day 6 (09/06/26 TUE) – Kubernetes Hands-On

- Created a local Kubernetes cluster using Docker Desktop.
- Learned core Kubernetes concepts including clusters, nodes, pods, deployments, replicas, and services.
- Understood the relationship between Docker containers and Kubernetes pods.
- Created an Nginx deployment and scaled it from one pod to three replicas.

- Observed how Kubernetes automatically manages and recreates pods to maintain deployment requirements.
 - Understood the role of Kubernetes in application scaling, self-healing, and container orchestration.
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What I Understand About SmartHub

- SmartHub follows a modular architecture where Web and Mobile applications communicate with centralised backend APIs.
 - Authentication, AI, RAG, Database, and Notification services operate as independent modules while integrating through backend services.
 - The RAG pipeline processes uploaded documents through extraction, chunking, embedding generation, vector storage, retrieval, and LLM-based answer generation.
 - PostgreSQL manages application data, while vector databases support semantic search of uploaded documents.
 - RabbitMQ enables asynchronous processing of notifications and background tasks.
 - Docker, Terraform, AWS, and Kubernetes together provide deployment, infrastructure management, scalability, and reliability.
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What I Learned Personally

- Gained a broader understanding of SmartHub beyond my original DevOps module.
- Learned the importance of understanding architectural decisions and technology trade-offs before implementation.
- Improved my ability to analyse alternative solutions and evaluate production-ready technologies.
- Developed a stronger systems-level understanding of how different modules interact within a complete product.
- Benefited from team discussions and knowledge-sharing sessions, which helped build a more detailed understanding of the project.